

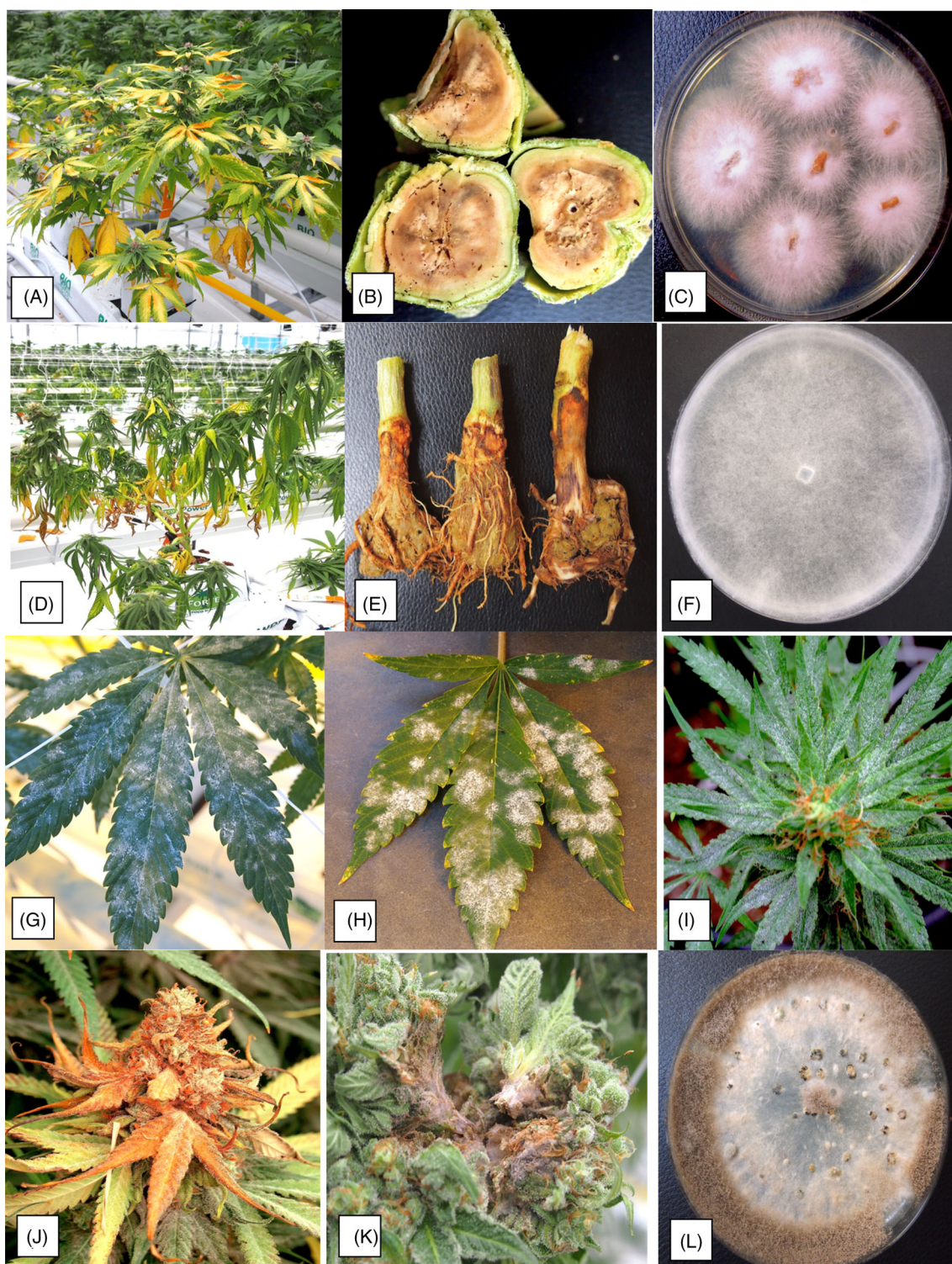


Figure 3. The pathogens that infect cannabis plants cause different symptoms. (A)–(C) *Fusarium* causes yellowing of foliage and internal stem necrosis. (D)–(F) *Pythium* causes wilt and crown necrosis. (G)–(I) *Golovinomyces* causes powdery mildew on leaves and inflorescences. (J)–(L) *Botrytis* causes bud rot. Respective pathogen cultures are shown in (C), (F) and (L).

bacterial pathogens affecting cannabis or hemp plants and the symptoms they cause are surprisingly few (Table S1); most appear to be saprophytes and incidental/secondary contaminants.^{7,9,35,44,45} Whether or not this is due to innate resistance in the plant or because environmental conditions have not supported bacterial infection or spread remains to be determined. It

is likely that members of the genera *Pseudomonas*, *Xanthomonas* and *Pectobacterium* will be the most commonly occurring bacterial pathogens in addition to those reported already. Viral and viroid pathogens of cannabis and hemp have been identified by molecular analysis that involves RT-PCR of RNA templates.^{22,23} In addition, next-generation sequencing (NGS) approaches are